

E2 Datawarehouse Design

Mérida Urban Intelligence

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Assignment: E2 Datawarehouse Design

Date: October 8, 2026

Course: Business Intelligence

Group: 9B – 2026Q3

Professor: Didier Omar Gamboa Angulo



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Mérida Urban Intelligence: Geospatial Data Warehouse

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Abstract—We integrated demographic, economic and road-safety records into a PostgreSQL/PostGIS warehouse for 483 urban AGEBS in Mérida. The study includes 921,771 residents, 54,995 establishments and 2,130 recorded traffic accidents. A modular ETL standardizes geographic keys, checks geometry and reconciles geographic and temporal totals. Pearson and Spearman correlations show a stronger association between business and accident density than between population and accident density. Global Moran’s I indicates positive spatial clustering, especially for population. Local and bivariate statistics describe clusters and relationships with neighboring areas. The results are exploratory because reference years differ, local tests are unadjusted and density indicators share the area denominator.

Index terms—Data warehouse, PostGIS, AGEB, road safety, territorial KPIs, spatial association.

I. Problem and data sources

A. Objective and authorized scope

Our objective was to compare areas within Mérida and explore how population, economic activity and road safety are geographically associated. Instead of combining municipal totals, the project integrates records at a common local unit and calculates comparable territorial indicators.

The instructor authorized the replacement of crime records with traffic accidents, as confirmed by the team. Accordingly, public safety is operationalized here as *road safety*. Accident indicators must not be interpreted as measures of crime or delinquency. This report communicates the implemented solution and findings; detailed execution instructions belong in the repository README.

B. Original sources and analytical variables

Four original INEGI archives support the analysis. They remain unchanged; the extraction module records their sizes and SHA-256 fingerprints.

Table 1: Source grain and retained information.

Source	Original grain and variables
Census 2020 [1]	Mixed locality, AGEB and block summaries. Population, age groups, active/inactive population and housing.
2020 cartography [2]	Official geographic features. Urban AGEB/locality polygons, identifiers and projected geometry.
DENUE May 2026 [3]	One establishment. ID, SCIAN activity/code, geographic identifiers and coordinates.
ATUS 2024 [4]	One georeferenced accident. ID, type, date/hour, fatalities, injuries and official point geometry.

The supplied archives are `ageb_mza_urbana_31_cpvt2020_csv.zip`, `31_yucatan.zip`, `denue_31_csv.zip` and `atus_2024_shp.zip`. The original DENUE metadata identifies release 05_2026 (May

2026), with identifier MEX-INEGLEEC2.05-DENUE-2026 and reference/modification date May 20, 2026. Original download dates were not recorded. Establishment registration dates do not define the snapshot date.

II. Geographic integration strategy

A. Selected unit and alternatives

The selected scope is entity 31, municipality 050 and locality 0001: the urban locality of Mérida. It contains 483 urban Basic Geostatistical Areas (AGEBs). Municipality-level analysis would hide internal variation. Blocks offer finer detail but more sparse counts and a heavier spatial integration workload. AGEBS provide a workable compromise between census detail, official polygons and neighborhood analysis.

The four-character AGEBS code is unique *inside this fixed locality*. It is not a national key. A future multi-locality model must use the complete geographic identifier or a surrogate key with a unique full-code constraint.

B. CRS, polygon checks and point assignment

Polygons are checked for valid, nonempty geometry, unique codes and positive area. Area is measured in the original projected INEGI CRS and converted from square meters to km². Geometry is then stored as WGS84 MultiPolygon (EPSG:4326) in PostGIS; area is not calculated from longitude/latitude degrees.

Census rows are filtered to urban AGEBS totals using locality codes, block code zero and the source’s total-AGEBS label. Geographic keys must match polygons one-to-one. DENUE uses normalized AGEBS codes; unmatched codes are tested using latitude/longitude points and audited. ATUS coordinates are also reconstructed and compared with official shapefile points. The largest observed difference was 0.0 m; official points remain the spatial authority. Points are reprojected into the polygon CRS and joined using `within`. Boundary or unmatched points receive no arbitrary nearest-AGEBS assignment.

III. ETL and warehouse architecture

A. Modular processing and validation

The Python workflow separates extraction, transformation, loading and analysis. Extraction reads the original archives and creates the source manifest. Transformation selects the study geography, cleans variables, integrates sources and exports validated datasets. Loading populates PostgreSQL/PostGIS, while analysis reads warehouse views and exports KPI tables, statistics and figures.

Census suppression marks `*` become missing values, never zero. Counts for AGEBS without assigned businesses or accidents become zero after aggregation. Duplicate source IDs, duplicate geographic keys, invalid coordinates, overlapping spatial assignments and infinite indicators raise errors instead of being silently accepted. Zero or missing denominators yield an unavailable rate.

Table 2: Observed integration and quality checks.

Check	Result
AGEBs and demographic/economic/accident rows	483 each
DENUE candidate records	55,059
DENUE integrated / excluded	54,995 / 64
Unmatched DENUE recovered by coordinates	0
Municipality ATUS records	2,271
ATUS integrated / excluded	2,130 / 141
Excluded ATUS points inside locality polygon	0
Temporal fact rows	1,988
Missing total-population values	0

The loader uses temporary staging tables, named columns and explicit type conversion within one transaction. Primary-key upserts support repeated loads of the current snapshot. Foreign keys connect each fact to the geographic dimension. Validation checks row counts, geometry, category totals and temporal reconciliation before commit. The team successfully repeated the warehouse load. This checks current-snapshot loading; it does not establish a historical versioning mechanism.

B. Dimensional model and grain

The implemented model has one shared geographic dimension and four fact tables. Three facts contain one row per AGEb for their respective reference snapshot; the fourth preserves accident aggregates by year, month and time-of-day period.

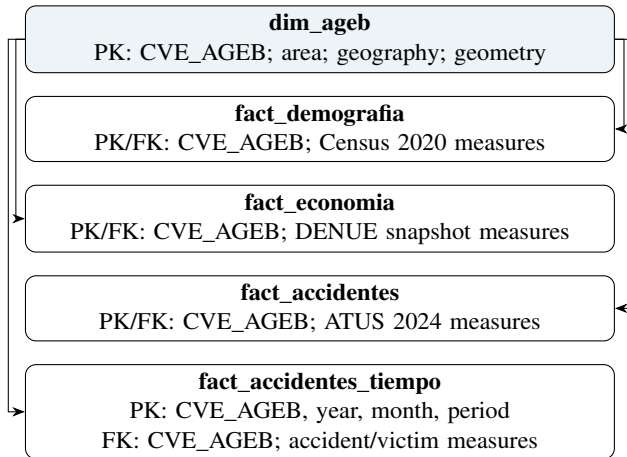


Figure 1: Each arrow connects the dimension to a fact through the AGEb key. Snapshot facts have one row per area; the temporal fact has many rows per area.

C. Concise dictionary and retained detail

Table 3: Main warehouse fields and meanings.

Field group	Meaning
CVE_AGEb	Four-character study-area key; PK/FK.
area_km2; geometry	Projected area; EPSG:4326 MultiPolygon, GiST indexed.
P0BT0T; age counts	Total residents; ages 0–14, 15–64 and 65+.
PEA, PE_INAC	Active and inactive population, used in activity rate.
Business counts	Total, commerce, retail, services and other establishments.
Dominant activity	Most frequent SCIAN activity; ties use ascending code.
Accident counts	Total events, nine observed type-code counts, deaths and injuries.
Temporal attributes	Year, month and period; event and victim totals.

Retail uses SCIAN 46; commerce includes 43 and 46. Services use sectors 48, 49, 51–56, 61, 62, 71, 72, 81 and 93. This project grouping includes public administration and should not be read as a universal sector definition.

Stored totals and denominators reproduce the required KPIs without rereading raw files. Accident types are retained in the AGEb fact; time is retained in a separate temporal fact. Their joint type-by-time distribution and individual events are not retained in this warehouse. Demographic and economic facts do not store historical versions. These are explicit limits of the current analytical grain.

IV. Territorial indicators and descriptive results

A. KPI definitions

For AGEb i , let A_i be area in km^2 , P_i population, B_i establishments and T_i accidents. The warehouse provides both counts and derived rates.

Table 4: Required indicator families and formulas.

Indicator	Calculation
Population density	P_i/A_i
Economically active rate	$100 PEA_i / (PEA_i + PE_INAC_i)$
Age distribution	$100 P_{i,g} / P_i$ for each age group g
Business total / density	$B_i; B_i/A_i$
Businesses per 1,000 residents	$1,000 B_i / P_i$
Retail / service densities	Category count divided by A_i
Dominant activity	Most frequent SCIAN activity
Accident total / density	$T_i; T_i/A_i$
Accidents per 1,000 residents	$1,000 T_i / P_i$
Accidents per establishment	T_i/B_i
Accident type / time	Type counts; month/period totals

Rates with nonpositive or missing denominators are null. Territorial totals are calculated by summing compatible measures; densities and rates must not be added or averaged without considering their denominators.

B. Study-area totals

The selected polygons cover 234.926 km² and contain 921,771 residents, 54,995 establishments and 2,130 accident records. Of the establishments, 18,686 are retail and 29,596 fall within the project's service grouping. Aggregate business intensity is 59.66 establishments per 1,000 residents, and recorded accidents amount to 2.31 per 1,000 residents. These ratios use Census 2020 population, not an estimate of 2024 exposure.

The highest observed population density is 18,064.50 residents/km² in AGEB 701A. The highest business density is 5,244.48 establishments/km² in AGEB 0510, where 2,932 establishments occupy a 0.559 km² polygon. This illustrates how a small denominator can magnify a density; it does not justify dropping that observation. AGEB 0879 has the highest accident density, 81.71 records/km².

The census age-count fields contain suppressed/missing values in 6 AGEBs for ages 0–14, 3 for ages 15–64 and 8 for ages 65+. Age shares are therefore calculated only where the required numerator and denominator are available. We do not fill suppressed values or claim complete age-group reconciliation.

C. Spatial and monthly distribution

Figure 2 shows recorded accident density, rather than raw totals. The unequal colors indicate territorial variation. This map supports comparison, but does not alone establish statistically significant clusters or identify causal factors.

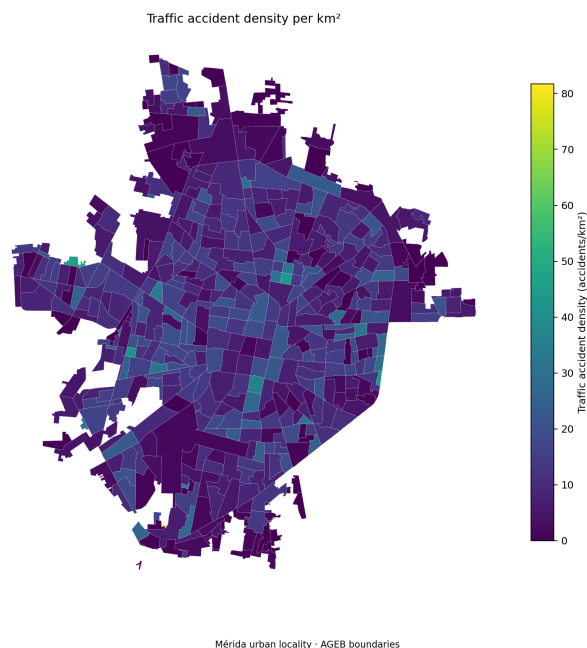


Figure 2: ATUS 2024 accidents per km² in the 483 selected urban AGEBs. Original project output.

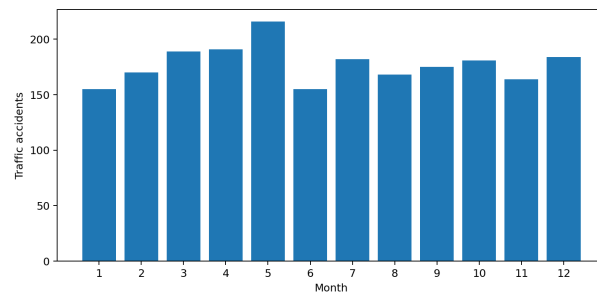


Figure 3: Monthly accident records integrated into the study area. Values sum to 2,130, not the municipality's 2,271.

May has the largest monthly count (216), with April (191) and March (189) also among the spring months; January has 155, the same as June. This is a description of one year. It does not establish recurring seasonality or a relationship with weather, traffic volume or holidays. Those explanations would require additional variables and multiple years.

Time periods are defined as early morning (hours 0–5), morning (6–11), afternoon (12–17) and night (18–23). The temporal fact supports comparisons by month and period. Unknown hours are retained as an unspecified period. Accident totals in this fact reconcile with the geographic fact.

V. Relationships and local spatial patterns

A. Correlation protocol and results

We compared three density pairs across all 483 AGEBs. Pearson measures linear association; Spearman measures rank association. Both use the original values, with no removal of extreme observations. Supplementary log1p plots preserve zeros and change only the visual scale.

Table 5: Within-AGEB associations; $n = 483$ in every row. P-values are nominal.

Pair	Method	Coefficient	p
Population/business	Pearson	0.1595	4.35×10^{-4}
Population/business	Spearman	0.4541	5.99×10^{-26}
Population/accident	Pearson	0.0920	0.0434
Population/accident	Spearman	0.1492	0.0010
Business/accident	Pearson	0.2387	1.10×10^{-7}
Business/accident	Spearman	0.4348	1.08×10^{-23}

Population and business densities have a moderate positive rank association, although their linear association is weak. Business and accident density likewise have a stronger rank association than a linear association. Population and accident density have the weakest association under both measures.

Figure 4 shows the business/accident pair. The log1p view makes low and intermediate values easier to inspect without hiding the highly concentrated business AGEB. The difference between Pearson and Spearman motivates attention to skewness and extreme values; it does not prove which mechanism generated the pattern. Statistical significance also does not imply a large effect.

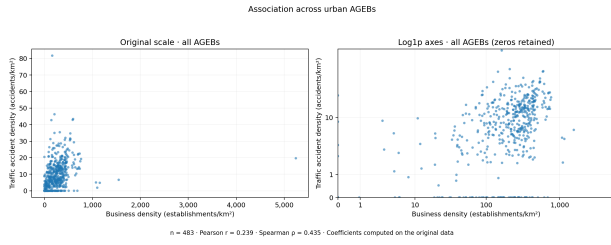


Figure 4: Business and accident density, original and log1p visual scales. Coefficients are calculated from original values.

All densities share the area denominator. Their associations may partly reflect area variation. Spatial dependence also challenges ordinary independent-observation p-values. Accordingly, we treat correlation results as descriptive/exploratory, not as causal evidence or a calibrated accident-risk model.

B. Neighborhood rule and LISA classification

The analysis uses Queen contiguity: polygons sharing an edge or vertex are neighbors. Weights are standardized by row. The geography has three connected components and one neighborless AGEb, 7039. All areas remain in global analysis; the island has zero spatial lag and receives a separate “No neighbors” label in local maps.

Local Moran statistics use 999 permutations, seed 42 and unadjusted $p < 0.05$. High/low refers to standardized values relative to the study-area distribution. High-high and low-low describe similar-value clusters; high-low and low-high describe spatial outliers. Non-significant areas are kept separate. Figures 5 and 6 use identical category colors, preventing a color from changing meaning between maps.

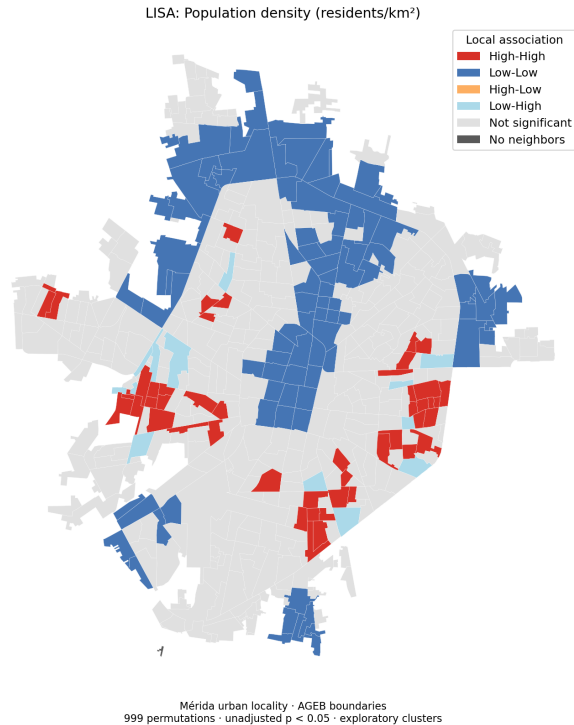


Figure 5: Population-density LISA: 49 high-high and 88 low-low areas.

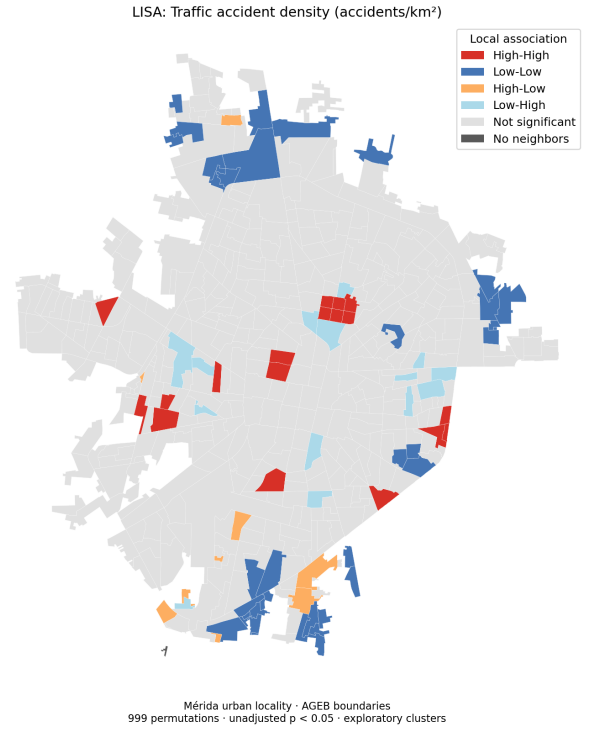


Figure 6: Accident-density LISA: 18 high-high and 36 low-low areas. Gray non-significant areas and the dark island remain distinct.

VI. Spatial statistics and main findings

A. Global and bivariate Moran

We calculated two univariate Moran statistics and one bivariate Moran statistic using the same geographic weights and 999 permutations [5]. The expected univariate value under randomization is $-1/(483 - 1) = -0.0021$.

Table 6: Spatial association on the retained geography.

Statistic	I	P_{sim}
Population-density Moran	0.4743	0.001
Accident-density Moran	0.0994	0.001
Business density / neighboring accident density	0.1218	0.001

Positive univariate values indicate that neighboring areas tend to have similar values. Population density shows a substantially stronger global pattern than accident density. Both differ from randomized arrangements under this particular neighborhood specification. A small permutation p-value does not make the accident clustering effect large, and 0.001 is the simulation resolution with 999 permutations.

The bivariate statistic relates business density in area i to the spatial lag of accident density in its neighbors. Its positive value indicates a geographic association between these two different variables. It is not the within-area business/accident Pearson correlation and does not establish that businesses cause nearby accidents.

B. Local results and practical reading

Table 7: LISA classification counts; totals include the island.

Category	Population	Accidents
High-high	49	18
Low-low	88	36
High-low	0	10
Low-high	13	14
Not significant	332	404
No neighbors	1	1
Total	483	483

Population has more classified similar-value clusters than accidents. Accident results are mostly non-significant (404 of 483 AGEBS); therefore, the entire density map should not be called a statistically significant hotspot map. The 18 high-high accident areas are candidates for further inspection, alongside the 10 high-low spatial outliers. Neither category alone defines an intervention priority without road exposure, severity, infrastructure and local validation.

VII. Limitations and interpretation cautions

A. Coverage and source quality

The geography covers the urban locality, not the whole municipality or metropolitan area. Excluded records remain in audits, and missing census values remain missing. Zero assigned events mean no matched records in this source and scope; they do not guarantee that no event occurred. ATUS is an administrative record system, so the observed records should not be treated as a complete count of every accident [4].

Population is from 2020, accidents from 2024, and DENU from May 2026. This temporal mismatch limits contemporaneous interpretation. Accident rates per resident describe territory, not individual travel risk: resident population is not traffic exposure. The verified DENU metadata and original archive fingerprints are preserved in the source inventory; original download dates were not recorded.

B. Statistical and modeling limits

LISA performs multiple local tests without a multiplicity adjustment. Its clusters are exploratory. No alternative-neighborhood, multiple-testing or island-exclusion sensitivity results are claimed. Disconnected components, spatial dependence, common area denominators, sparse counts and extreme densities can affect interpretation. Findings may change with geographic aggregation or another boundary system.

The warehouse preserves sufficient aggregates for the selected metrics but does not retain raw event-level records, historical demographic/economic snapshots or a complete type-by-time accident cube. Expansion beyond Mérida requires globally unique geographic keys. The standalone HTML dashboard is an exported snapshot rather than a live database connection; its filters, charts and maps support review of the same results, not new inference.

VIII. Conclusion

The project provides a common geographic warehouse for comparing Mérida’s urban areas. Validated ETL links census summaries, establishments and georeferenced accidents, while stored measures support demographic, economic and road-safety KPIs. Business density is more closely associated with accident density than population density is, especially by rank. Population shows stronger spatial clustering, while most accident LISA tests are non-significant. These findings support further territorial investigation but do not establish causal relationships. Additional exposure measures, aligned reference periods and sensitivity analysis would strengthen future decisions.

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